

Geonwoo Kim

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“There is no failure, only growth.”

Education

Inha University, Department of Artificial Intelligence

B.S. IN ARTIFICIAL INTELLIGENCE

- GPA: **4.16 / 4.5**.

Incheon, S. Korea

Mar. 2021 – Aug. 2027 (Expected)

Research Interests

Generative Models, Multimodal Learning, World Models

Research Experience

DAVIAN Lab | Kim Jaechul Graduate School of AI, KAIST

UNDERGRADUATE RESEARCH INTERN

- Researched **multi-view conditioned video generation**, integrating 3D-grounded data synthesis and geometric conditioning.
- Co-led the research as a **co-first author** and submitted the paper to **ICLR 2027**; advised by Professor Jaegul Choo.

Seoul, S. Korea

Jan. 2026 – Present

Multimodal AI Lab | Inha University

UNDERGRADUATE RESEARCHER

- Reproduced and compared **training-free subject-driven image editing** methods using reference images, target images, and masks.
- Extended masked editing to preserve subject identity and background context; advised by Professor Pilhyeon Lee.

Incheon, S. Korea

Dec. 2024 – Dec. 2025

Publications

A Paper on Multi-view Conditioned Video Generation

CO-FIRST AUTHOR

- Developed 3D-grounded data synthesis and geometric conditioning for **subject-consistent video generation** across viewpoints.

ICLR 2027, under review

2026

Project Experience

Action-Conditioned Robot World Model

TEAM LEADER

- Led an **action-conditioned video generation** pipeline predicting future robot observations from an initial image and an action sequence.
- Adapted **Cosmos-Predict2.5-2B** with **LoRA**, an inverse-dynamics auxiliary loss, and model soup, integrating training through final inference.
- Reduced the public leaderboard score by **66.1% over the official baseline**, winning the **Top Excellence Award (2nd, graduate track)**.

Code / Slides

Jul. 2026 – Aug. 2026

LG Aimers 9th: Pitch Control Prediction

TEAM LEADER

- Engineered historical pitching features to predict **KBO pitch-control success probabilities**.
- Applied residual modeling and segment-wise calibration to a **publicly available CatBoost ensemble backbone**.
- Achieved **2.09 times the official baseline's score** on the public leaderboard, ranking **88th**.

Code

2026

Agent Action Prediction

TEAM LEADER

- Built a **14-class next-action predictor** from agent interaction history with session-level validation to prevent leakage.
- Optimized encoder ensembles through **vocabulary pruning** and conditional inference under model-size and runtime constraints.
- Achieved a final leaderboard score of **0.78837** after controlled experiments on input serialization, augmentation, and ensemble selection.

Code

Jul. 2026

COSFLOW: AI Automation Solution for Cosmetic OEM Companies

Website

CO-FOUNDING MEMBER / AI RESEARCHER

Jun. 2025 – Sep. 2025

- Developed an **AI Agent solution** streamlining cosmetic ingredient compliance workflows for OEM clients.
- Built an **LLM-integrated pipeline** that standardizes ingredient names, queries multi-country databases, and generates structured compliance reports (PDF).
- Implemented **vector database-based ingredient similarity search** and **GPT-4.1-driven report generation**, including ingredient cards and dashboards.

Text-Guided Image Colorization (Inha AI Challenge)

Code

TEAM LEADER

Jun. 2025 – Aug. 2025

- Led an AI competition project focused on **text-guided colorization** of grayscale images.
- Built a candidate-generation pipeline combining **DDColor** and **FLUX** for text-conditioned colorization.
- Designed **CLIP-based candidate selection** to choose the generated image most aligned with the text description.
- Achieved the **BK21 Director's Award (Excellence)** with a final score of **0.6178**.

AI-Generated vs. Human-Written Text Classification Challenge

Seoul, S. Korea

TEAM LEADER

Jun. 2025 – Jul. 2025

- Developed an AI model distinguishing **human-written and LLM-generated Korean texts**, despite full-text-level labeling.
- Improved paragraph-level discrimination via **LLM-based data augmentation** and **ensemble modeling**.
- Used **RoBERTa-Large, BERT-base, and KoBigBird** embeddings with **3:1 undersampling** to address imbalance.
- Achieved **ROC-AUC 0.906** in the classification challenge.

Enterprise-tailored AI Research & Report Generation Agent

Incheon, S. Korea

CO-FOUNDING MEMBER / AI RESEARCHER

Mar. 2025 – Jun. 2025

- Developed an **AI Agent system integrating RAG retrieval**, vector database search, and template-based report automation.
- Built a **LangChain-powered CSV Analysis Agent** supporting data queries, visualization, and automated summaries in Streamlit.
- Designed an **LLM pipeline** for domain-adaptive generation and reliability evaluation of external sources.
- Secured government funding through the **Korean Pre-Startup Package** and **Lab-based Startup Leading University Program**.

Awards & Honors

2026.09	Top Excellence Award (2nd, graduate track) , Robot World Model AI Challenge	Inha Univ.
2026.05	Excellence Award (3rd) , Inter Campus AI Challenge 2026	Upstage
2025.09	Director's Award (Excellence, 3rd) , Incheon Startup Competition	Incheon
2025.08	BK21 Director's Award (Excellence, 3rd) , Text-Guided Colorization	Inha Univ.
2025.06	Mayor's Award (Outstanding Idea, 3rd) , Incheon Startup Competition	Incheon
2025.05	Grand Prize (President's Award, 1st) , MOONG IT Hackathon	Inha Univ.
2025.03	Grand Prize (1st) , Launch Pad Startup Competition	Inha Univ.
2024.11	1st Place , GDG Ideathon & Demo Day	Inha Univ.
2022.09	Dean's Award (1st) , Academic Excellence, AI Department	Inha Univ.
2021	Academic Excellence Scholarship , Academic achievement	Inha Univ.

Skills

Programming	Python, C++
Deep Learning	PyTorch, TensorFlow
RAG & Agents	LangChain, LangGraph, FAISS, Streamlit, MCP
3D & Design	Blender, Figma
Tools	Linux, Docker, Git, GitHub, SSH
Libraries	NumPy, Pandas, Matplotlib, Scikit-learn

Extracurricular Activity

AI Society: SINSA VICE PRESIDENT - Managed a university AI society under faculty supervision, overseeing programs in Python, Data Analysis, Deep Learning, and Paper Study. - Organized internal AI competitions and structured study curricula, contributing to national awards in SW-centered University AI competitions.	<i>Incheon, S. Korea</i> <i>Jun. 2025 – Jun. 2026</i>
RAG/Agent Study STUDY LEADER - Led a study group on RAG and LLM-based Agents using LangChain, LangGraph, and MCP. - Combined theoretical exploration with hands-on multimodal RAG/Agent implementation and discussions of research and industrial applications.	<i>Incheon, S. Korea</i> <i>Aug. 2025 – Present</i>
Bishop-Deep Learning Study STUDY MEMBER Participated in a year-long study of Bishop's <i>Deep Learning: Foundations and Concepts</i>, presenting CNNs, diffusion models, and Transformers through weekly peer teaching.	<i>Incheon, S. Korea</i> <i>Mar. 2025 – Present</i>
Start-up AI Agent Lab: UPFLOW CO-FOUNDING MEMBER / AI RESEARCHER - Led an AI Agent startup supported by the Pre-Startup Package and Lab-based Startup Program. - Developed enterprise AI Agent and MCP-based automation for consulting and cosmetic OEM clients, gaining end-to-end experience from model development to commercialization.	<i>Incheon, S. Korea</i> <i>Feb. 2025 – Oct. 2025</i>
CS231n Study (Stanford Course) STUDY MEMBER Participated in weekly discussions of Stanford CS231n, presenting CNN architectures and applications in computer vision.	<i>Incheon, S. Korea</i> <i>Mar. 2025 – Jun. 2025</i>

Teaching

Environmental Data Analysis Mentoring MENTOR - Mentored middle and high school students in environmental data analysis using KoNLPy, stopwords removal, and Kkma to analyze carbon-neutrality issues. - Implemented a recycling-status dashboard and object detection model for recyclable-item classification.	<i>Incheon Metropolitan Office of Education</i> <i>May. 2025 – Jun. 2025</i>
Digital Green Workshop MENTOR Mentored high school students in AI-based environmental analysis, guiding problem definition and blockchain incentive solutions with Inha University and the Incheon Metropolitan Office of Education.	<i>Inha University</i> <i>Oct. 2024</i>
EBS 1318 On-Coaching Program MENTOR Provided individual mathematics and science tutoring in equations, functions, and set theory, emphasizing conceptual understanding.	<i>Siheung, S. Korea</i> <i>Jul. 2024 – Present</i>
Hwasung Mathematics Academy TEACHING ASSISTANT - Assisted curriculum design and personalized learning plans; taught probability, statistics, and calculus for college entrance preparation. - Communicated with students and parents to coordinate academic goals and progress.	<i>Yongin, S. Korea</i> <i>Mar. 2021 – Feb. 2022</i>

Certifications

ADsP ADVANCED DATA ANALYTICS SEMI-PROFESSIONAL	<i>Korea Data Agency</i> <i>Mar. 22, 2024</i>
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